

MATH 475: Honours Partial Differential Equations

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Abstract

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1 Introduction

Taught by Adam Oberman. Mondays and Wednesdays from 14:35 to 15:55 in Stewart Biology Building N2/2. Course grading is divided as follows:

- 10% 5-6 assignments.
- 10% attendance and participation.
- 10% 3-5 quizzes.
- 20% midterm.
- 50% final exam.

2 Basic Definitions and Classification

2.1 Ways to Write a PDE

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Many of the classical equations are built from a single operator, the Laplacian

$$\Delta u = \sum_{i=1}^n \frac{\partial^2 u}{\partial x_i^2}. \quad (1)$$

Equation	Form	Type
Heat	$u_t = \Delta u, \quad t \geq 0$	parabolic
Poisson	$\Delta u = f$	elliptic (stationary)
Laplace	$\Delta u = 0$	elliptic (stationary)
Wave	$u_{tt} = \Delta u$	hyperbolic

Not every PDE fits this pattern. Nonlinear examples include Burgers' equation

$$u_t + uu_x = 0, \quad (2)$$

and the Navier-Stokes equations.

2.2 Order and Solutions

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Definition 1 (Order). *The order of a PDE is the order of the highest derivative appearing in it.*

For example $u_{xx} = 0$ is second order, and $u_t + uu_x = 0$ is first order.

Definition 2 (Solution). *Let the PDE be written as $F(u) = 0$ and let k be its order. A function u is a solution if*

- (1) $u \in C^k$, that is u is k times continuously differentiable, and
- (2) $F(u) = 0$ at every point of the domain.

Example (Transport equation). Let $u = u(x, t)$ and $F(u_t, u_x) = u_t + cu_x = 0$ with $c \in \mathbb{R}$. Let Φ be any differentiable function and set $u(x, t) = \Phi(x - ct)$. Then

$$u_x = \Phi'(x - ct), \quad u_t = -c\Phi'(x - ct), \quad (3)$$

so $u_t + cu_x = 0$. The whole family Φ solves the equation.

Example (Heat equation). Let $u_t = c^2 u_{xx}$ and take the ansatz $u(x, t) = e^{-c^2 k^2 t} \sin(kx)$ for an arbitrary constant k . Then

$$u_t = -c^2 k^2 u, \quad u_x = k e^{-c^2 k^2 t} \cos(kx), \quad u_{xx} = -k^2 e^{-c^2 k^2 t} \sin(kx) = -k^2 u, \quad (4)$$

so $u_t = c^2 u_{xx}$.

Example (Wave equation). Let $u_{tt} = c^2 u_{xx}$ and take $u(x, t) = \cos(kct) \sin(kx)$. Then

$$u_{xx} = -k^2 u, \quad u_{tt} = -c^2 k^2 u, \quad (5)$$

so $u_{tt} = c^2 u_{xx}$.

2.3 Linear and Nonlinear

Definition 3 (Linear PDE). A PDE $F(u, u_x, x) = 0$ is linear if it can be rewritten as

$$L(u) = f(x), \quad (6)$$

where L is linear in u and its derivatives and f depends on x only.

Example. Take $F(u, u_x, x) = 3u + u_x - x^2$. Setting $L(u, u_x) = 3u + u_x$ and $f(x) = x^2$ gives

$$F(u, u_x, x) = 0 \iff L(u, u_x) = f(x), \quad (7)$$

so the equation is linear. The same definition works in higher dimensions.

Definition 4 (Homogeneous). A linear PDE $L(u) = f(x)$ is homogeneous if $f \equiv 0$, and inhomogeneous otherwise.

Example. $u_t + au_x = x^3$ is linear with $L(u) = u_t + au_x$ and $f(x) = x^3$, so it is inhomogeneous.

Example. $u_t + uu_x = f(x)$ is nonlinear, because the term uu_x is not linear in u .

2.4 Principle of Superposition

Principle 1 (Superposition). Let $L(u) = 0$ be linear and homogeneous. If u_1 and u_2 are solutions, then so is $u_1 + cu_2$ for every $c \in \mathbb{R}$.

Proof. By linearity of L ,

$$L(u_1 + cu_2) = L(u_1) + cL(u_2) = 0 + c \cdot 0 = 0. \quad \square \quad (8)$$

Example. For $u_t = c^2 u_{xx}$, both

$$u_1(x, t) = e^{-c^2 k^2 t} \sin(kx) \quad \text{and} \quad u_2(x, t) = e^{-c^2 t} \sin(x) \quad (9)$$

are solutions, so $u(x, t) = a_1 u_1(x, t) + a_2 u_2(x, t)$ is a solution for any $a_1, a_2 \in \mathbb{R}$.

¹

3 General Solutions

3.1 Solving by Direct Integration

Example. Solve $u_x = 0$ for $u(x, y)$ on $\mathbb{R}^2$. The solution is $u(x, y) = f(y)$ for an arbitrary f .

Example. Solve $u_{xx} = 0$. Set $v(x, y) = u_x$. Then $v_x = 0$, so by the previous example $v(x, y) = f(y)$, hence $u_x = f(y)$. Fixing y and integrating in x gives

$$u(x, y) = f(y)x + g(y). \quad (10)$$

Check: $u_x = f(y)$ and $u_{xx} = 0$.

Example. Solve $u_{xy} = 0$. Set $v(x, y) = u_x$. Then $v_y = 0$, so $v(x, y) = f(x)$, hence $u_x = f(x)$. Fixing y and integrating in x gives

$$u(x, y) = F(x) + g(y), \quad F' = f. \quad (11)$$

Check: $u_x = F'(x)$ depends on x only, so $u_{xy} = 0$.

Remark 1. In each case the general solution carries arbitrary functions rather than arbitrary constants. An ODE of order k has k free constants; a PDE has free functions of the remaining variables.

3.2 Auxiliary Conditions

Auxiliary conditions means boundary conditions. A typical problem on a domain $\Omega \subset \mathbb{R}^d$ reads

$$\Delta u = f(x) \quad \text{in } \Omega, \quad u(x) = g(x) \quad \text{on } \partial\Omega. \quad (12)$$

These conditions are what cut the family of general solutions down to one.

3.3 Operator Viewpoint

Write a linear PDE as

$$Lu = f, \quad (13)$$

where L is the operator, u is the unknown and f is the input. The linear algebra analogue is

$$Mx = b, \quad x, b \in \mathbb{R}^d, \quad x = M^{-1}b, \quad (14)$$

so solving a linear PDE is the infinite-dimensional version of inverting M .

¹80% chance that what we have written so far is not on the final.

4 Potential Theory

4.1 Newton's Problem

Place a point mass M at the origin and a test mass m at $x \in \mathbb{R}^3 \setminus \{0\}$. The gravitational force on m has magnitude $GmM/\|x\|^2$ and points back toward the origin:

$$\vec{F}(x) = -\frac{GmM}{\|x\|^2} \cdot \frac{x}{\|x\|} = -\frac{GmM}{\|x\|^3} x. \quad (15)$$

4.2 Toolkit

Lemma 1 (Derivative of distance is direction). *Let $r(x) = \|x\|$ with $x = (x_1, x_2, x_3) \in \mathbb{R}^3$. Then for $x \neq 0$,*

$$\frac{\partial r}{\partial x_i} = \frac{x_i}{r}, \quad \nabla r(x) = \frac{x}{r} = \hat{x}. \quad (16)$$

Proof. Start from $r^2 = x_1^2 + x_2^2 + x_3^2$ and differentiate both sides with respect to x_i :

$$2r \frac{\partial r}{\partial x_i} = 2x_i \implies \frac{\partial r}{\partial x_i} = \frac{x_i}{r}. \quad (17)$$

This requires $r \neq 0$, so the identity fails at the origin. $\square$

Remark 2. ∇r is the unit vector pointing radially outward, so $\|\nabla r\| = 1$ everywhere it is defined.

4.3 Radial Chain Rule

Proposition 1. *Let $f : (0, \infty) \rightarrow \mathbb{R}$ be differentiable and set $g(x) = f(r(x)) = f(\|x\|)$. Then for $x \neq 0$,*

$$\nabla g(x) = f'(r) \nabla r(x) = f'(r) \frac{x}{r}. \quad (18)$$

Proof. Componentwise, by the chain rule,

$$\frac{\partial g}{\partial x_i} = f'(r(x)) \frac{\partial r}{\partial x_i} = f'(r) \frac{x_i}{r}. \quad (19)$$

$\square$

So the gradient of any radial function points radially, and f' alone controls the magnitude.

Example (Gradient refresher, non-radial). *For $g(x) = x_1 x_2 + x_3^2$,*

$$\nabla g(x) = \left(\frac{\partial g}{\partial x_1}, \frac{\partial g}{\partial x_2}, \frac{\partial g}{\partial x_3} \right) = (x_2, x_1, 2x_3). \quad (20)$$

Remark 3. *The chain rule result above, $\nabla g(x) = f'(r(x)) \nabla r(x) = f'(r) \frac{x}{r}$, is the tool we use whenever a function depends only on distance from the origin.*

4.4 The Laplacian

Everything below works in $\mathbb{R}^d$ for any $d \geq 1$. Write $x = (x_1, \dots, x_d)$.

First form: sum of second partials

Definition 5 (Laplacian).

$$\Delta = \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2}, \quad \Delta f(x) = \sum_{i=1}^d \frac{\partial^2 f}{\partial x_i^2}(x). \quad (21)$$

In $d = 3$ this reads

$$\Delta f(x) = \frac{\partial^2 f}{\partial x_1^2}(x) + \frac{\partial^2 f}{\partial x_2^2}(x) + \frac{\partial^2 f}{\partial x_3^2}(x). \quad (22)$$

Example. Let $f(x_1, x_2, x_3) = x_1^2 - x_2^2$. Then

$$\frac{\partial^2 f}{\partial x_1^2} = 2, \quad \frac{\partial^2 f}{\partial x_2^2} = -2, \quad \frac{\partial^2 f}{\partial x_3^2} = 0, \quad (23)$$

so $\Delta f = 2 - 2 + 0 = 0$. A function with $\Delta f = 0$ is called harmonic.

Second form: divergence of the gradient

Definition 6 (Divergence). For a vector field $\vec{v}(x) = (v^1(x), \dots, v^d(x))$, where the superscripts label components and are not powers,

$$\nabla \cdot \vec{v} = \operatorname{div} \vec{v} = \sum_{i=1}^d \frac{\partial v^i}{\partial x_i}(x). \quad (24)$$

Applying this to the gradient field $\vec{v} = \nabla f$, whose i -th component is $\partial f / \partial x_i$, gives

$$\nabla \cdot (\nabla f) = \sum_{i=1}^d \frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_i} \right) = \sum_{i=1}^d \frac{\partial^2 f}{\partial x_i^2} = \Delta f. \quad (25)$$

Proposition 2.

$$\Delta = \nabla \cdot \nabla = \operatorname{div} \circ \operatorname{grad}. \quad (26)$$

Remark 4. This second form is the useful one for radial functions: combining it with the radial chain rule turns Δ into an ODE in r .

4.5 Harmonic Radial Functions

Proposition 3 (Radial Laplacian). For $g(x) = f(r)$, $r = \|x\|$ in $\mathbb{R}^d$, $x \neq 0$:

$$\Delta g = f''(r) + \frac{d-1}{r} f'(r). \quad (27)$$

Solve $\Delta f = 0$ in $d = 3$ with the ansatz $f(r) = r^a$:

$$\Delta f = a(a-1)r^{a-2} + \frac{2}{r}ar^{a-1} = a(a+1)r^{a-2}, \quad (28)$$

which vanishes for $r > 0$ iff $a = 0$ or $a = -1$. So

$$g(x) = A + \frac{B}{\|x\|}. \quad (29)$$

Remark 5. $1/\|x\|$ is the Newtonian potential: $\nabla(1/r) = -x/\|x\|^3$, the inverse-square field. Harmonic away from the origin only.

5 First Order PDEs and the Solution Operator

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5.1 The Underlying ODE

Start from the ODE

$$x'(s) = v(x(s), s), \quad (30)$$

where v is a velocity field. Assume $v(x, s)$ is smooth. By the existence and uniqueness theorem there is a unique trajectory passing through every point.

5.2 The Solution Operator

Definition 7 (Solution operator). *Given times t_0 and t_1 , the solution operator is the map*

$$S_{t_0 \rightarrow t_1}(x_0) = x(t_1), \quad (31)$$

where $x(\cdot)$ solves the ODE with initial condition $x(t_0) = x_0$.

Proposition 4 (Identities). 1. $S_{t \rightarrow t}(x) = x$, that is $S_{t \rightarrow t} = \text{Id}$.

2. $S_{t_1 \rightarrow t_2} \circ S_{t_0 \rightarrow t_1} = S_{t_0 \rightarrow t_2}$.

3. $S_{t_1 \rightarrow t_0} = (S_{t_0 \rightarrow t_1})^{-1}$: running time backwards inverts the operator.

5.3 Examples of Trajectories

Example. The curve $\vec{x}(s) = (x(s), y(s)) = (3e^{s-2}, 4e^{s-2})$ satisfies $\vec{x}'(s) = \vec{x}(s)$, so it is a trajectory of the velocity field $\vec{v}(x, y) = (x, y)$. It passes through $\vec{x}(2) = (3, 4)$ and $\vec{x}(5) = (3e^3, 4e^3)$.

Example. The curve $\vec{x}(s) = (3s+5, 5s+6)$ has constant velocity $\vec{v}(s) = \vec{x}'(s) = (3, 5)$, with $\vec{x}(0) = (5, 6)$ and $\vec{x}(2) = (11, 16)$.

Example (Decoupled linear field). Take $\vec{v}(x, y) = (c_1x, c_2y)$ for constants c_1, c_2 , so the ODE is

$$\begin{cases} x'(s) = c_1x(s) \\ y'(s) = c_2y(s) \end{cases} \quad \begin{cases} x(0) = x_0 \\ y(0) = y_0. \end{cases} \quad (32)$$

The solution is $x(s) = x_0 e^{c_1 s}$, $y(s) = y_0 e^{c_2 s}$. Check: $x'(s) = c_1 x_0 e^{c_1 s} = c_1 x(s)$, and similarly for y .

Example (Rotation field). Take $\vec{v}(x, y) = (-y, x)$. A circle of radius r , $\vec{x}(s) = (r \cos s, r \sin s)$, has $\vec{x}'(s) = (-r \sin s, r \cos s) = (-y, x)$, so circles about the origin are trajectories. The general solution of

$$\begin{cases} x' = -y \\ y' = x \end{cases} \quad \begin{cases} x(0) = x_0 \\ y(0) = y_0 \end{cases} \quad (33)$$

is

$$x(s) = x_0 \cos s - y_0 \sin s, \quad y(s) = x_0 \sin s + y_0 \cos s. \quad (34)$$

The circle above is the case $x_0 = r$, $y_0 = 0$.

5.4 Constant Coefficient Transport

Consider

$$au_x + bu_y = 0, \quad u = u(x, y), \quad a, b \in \mathbb{R}. \quad (35)$$

Proposition 5 (Constancy along characteristics). Let $\vec{v} = (a, b)$ and let $\vec{x}(s) = (x(s), y(s))$ solve $\vec{x}'(s) = \vec{v}$, that is $x'(s) = a$ and $y'(s) = b$. Then $u(\vec{x}(s))$ is constant in s .

Proof. By the chain rule,

$$\frac{d}{ds} u(x(s), y(s)) = u_x x'(s) + u_y y'(s) = au_x + bu_y = 0. \quad \square \quad (36)$$

The trajectories of $\vec{v}$ are called *characteristics*. Since u is constant along each one, its value is determined by wherever the characteristic meets the data.

Proposition 6 (General solution). For any differentiable f ,

$$u(x, y) = f(-bx + ay) \quad (37)$$

solves $au_x + bu_y = 0$.

Proof. $u_x = -b f'(-bx + ay)$ and $u_y = a f'(-bx + ay)$, so $au_x + bu_y = (-ab + ab) f'(-bx + ay) = 0$. $\square$

Remark 6. The argument $-bx + ay$ is constant exactly along the lines of direction (a, b) , which are the characteristics. When $b = 0$ the equation reduces to $au_x = 0$ and the formula gives $u(x, y) = f(ay)$, a function of y alone, as expected.

5.5 Adding Data

Example. Solve

$$\begin{cases} 3u_x + 2u_y = 0 & (PDE) \\ u(x, 0) = x^3 & (IC), \end{cases} \quad (38)$$

where the data is given on the line $\Gamma = \{(x, y) \mid y = 0\}$.

Method 1 (Via the general solution). With $a = 3$, $b = 2$, the general solution is $u(x, y) = f(-2x + 3y)$. On $y = 0$ this reads $u(x, 0) = f(-2x) = x^3$. Setting $z = -2x$, so $x = -z/2$, gives $f(z) = -z^3/8$. Therefore

$$u(x, y) = f(-2x + 3y) = -\frac{(-2x + 3y)^3}{8} = \frac{(2x - 3y)^3}{8}. \quad (39)$$

Check: $u(x, 0) = (2x)^3/8 = x^3$, and $3u_x + 2u_y = \frac{3}{8} \cdot 3(2x - 3y)^2 \cdot 2 + \frac{2}{8} \cdot 3(2x - 3y)^2 \cdot (-3) = 0$.

Method 2 (Via characteristics). (1) u is constant along curves with $x'(s) = 3$, $y'(s) = 2$.

(2) These are the lines $x(s) = 3s + x_0$, $y(s) = 2s + y_0$.

(3) Given a target point (x, y) , find the point $(x_0, 0)$ on Γ whose characteristic reaches it. From $y = 2s + 0$ we get $s = y/2$, and then $x = 3s + x_0$ gives $x_0 = x - 3y/2$.

(4) Since u is constant along the characteristic, $u(x, y) = u(x_0, 0) = x_0^3$:

$$u(x, y) = \left(x - \frac{3y}{2}\right)^3 = \frac{(2x - 3y)^3}{8}. \quad (40)$$

Both methods agree.

6 Misc

A Appendix

B Solutions

C Useful Links